

Kazakh-British Technical University

Databases

Laboratory work № 3

DML.

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1. Generate and insert 200 rows

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including the 'airport' table. The central pane shows the SQL query editor with the following code:

```
1 INSERT INTO Airport (airport_name, country, state, city, created_at, updated_at)
2 SELECT
3   'Airport_' || i,
4   'Country_' || (random()*50)::int,
5   'State_' || (random()*50)::int,
6   'City_' || (random()*50)::int,
7   NOW(),
8   NOW()
9 FROM generate_series(1,200) AS s(i);
```

The bottom pane shows the 'Data Output' tab with the message: "INSERT 0 200" and "Query returned successfully in 59 msec." The status bar at the bottom indicates "Total rows: Query complete 00:00:00.059".

2. Add a new airline named "KazAir" based in "Kazakhstan" to the airline table.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including the 'airline_info' table. The central pane shows the SQL query editor with the following code:

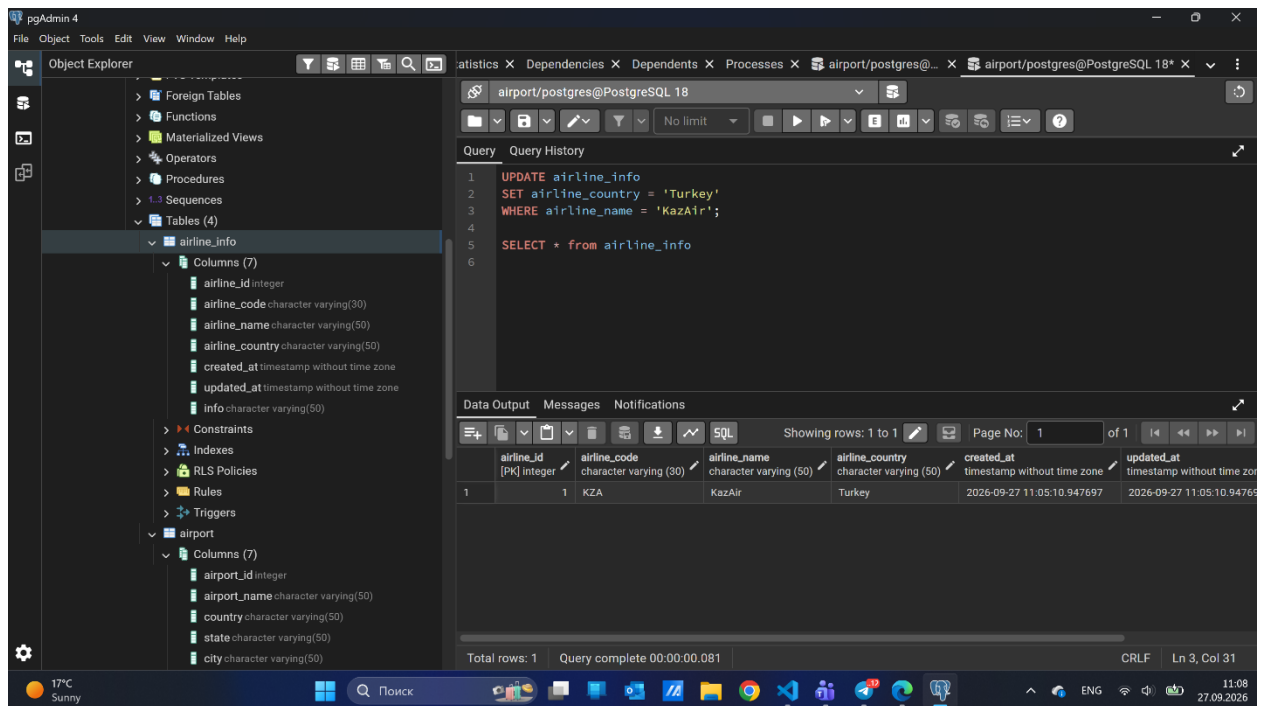
```
1 INSERT INTO airline_info(airline_code, airline_name, airline_country, created_at, updated_at)
2 VALUES ('KZA', 'KazAir', 'Kazakhstan', NOW(), NOW());
3
4 SELECT * from airline_info
5
6
```

The bottom pane shows the 'Data Output' tab with a table of results:

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
1	KZA	KazAir	Kazakhstan	2026-09-27 11:05:10.947697	2026-09-27 11:05:10.947697

The status bar at the bottom indicates "Total rows: 1" and "Query complete 00:00:00.079".

3. Update the airline country "KazAir" to "Turkey".



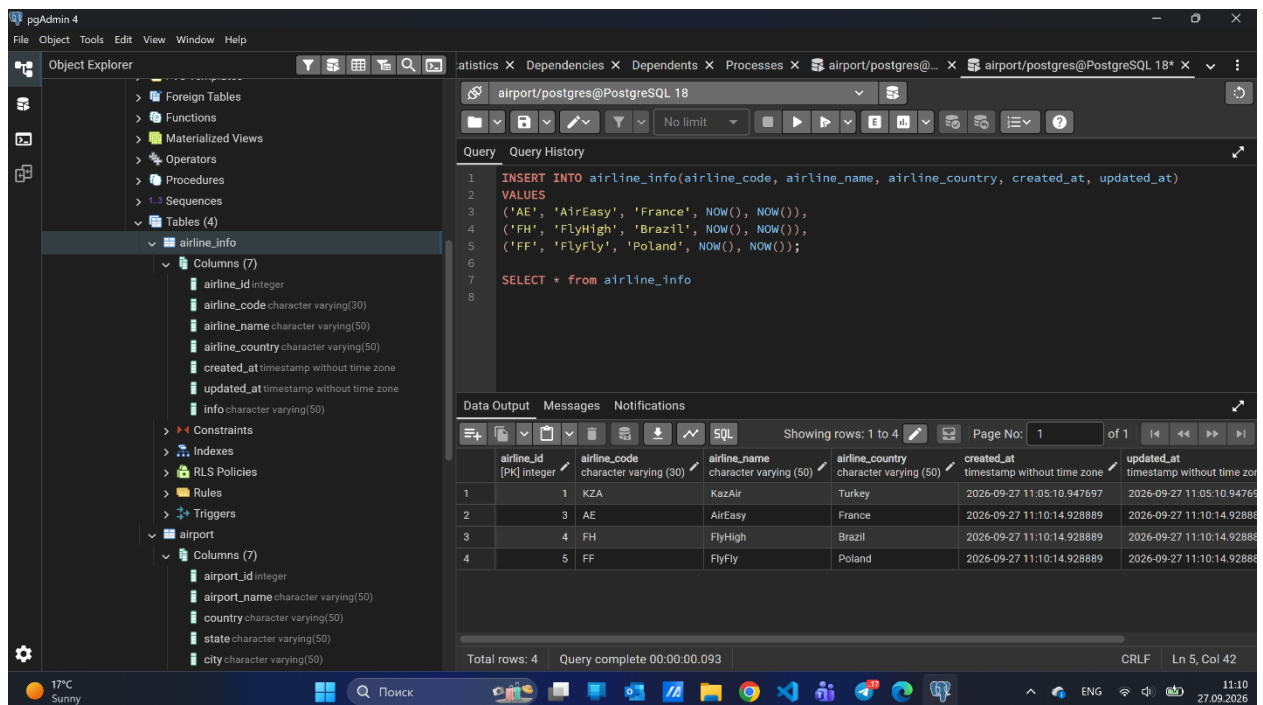
The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, including the 'airline_info' table with columns: airline_id, airline_code, airline_name, airline_country, created_at, updated_at, and info. The central pane shows a SQL query to update the 'airline_country' of 'KazAir' to 'Turkey'. The bottom pane shows the 'Data Output' table with one row.

```
UPDATE airline_info
SET airline_country = 'Turkey'
WHERE airline_name = 'KazAir';

SELECT * from airline_info
```

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
1	KZA	KazAir	Turkey	2026-09-27 11:05:10.947697	2026-09-27 11:05:10.947697

4. Add three airlines at once: "AirEasy" in "France", "FlyHigh" in "Brazil" and "FlyFly" in "Poland".



The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, including the 'airline_info' table. The central pane shows a SQL query to insert three new airlines: 'AirEasy' in France, 'FlyHigh' in Brazil, and 'FlyFly' in Poland. The bottom pane shows the 'Data Output' table with four rows.

```
INSERT INTO airline_info(airline_code, airline_name, airline_country, created_at, updated_at)
VALUES
('AE', 'AirEasy', 'France', NOW(), NOW()),
('FH', 'FlyHigh', 'Brazil', NOW(), NOW()),
('FF', 'FlyFly', 'Poland', NOW(), NOW());

SELECT * from airline_info
```

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
1	KZA	KazAir	Turkey	2026-09-27 11:05:10.947697	2026-09-27 11:05:10.947697
2	AE	AirEasy	France	2026-09-27 11:10:14.928889	2026-09-27 11:10:14.928889
3	FH	FlyHigh	Brazil	2026-09-27 11:10:14.928889	2026-09-27 11:10:14.928889
4	FF	FlyFly	Poland	2026-09-27 11:10:14.928889	2026-09-27 11:10:14.928889

5. Delete all flights whose arrival in 2024 year.

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including the 'flight' table. The main pane shows a SQL query to delete flights scheduled for 2024. The 'Data Output' tab displays the results of the query, showing 7 rows of flight data.

```
DELETE FROM flight
WHERE EXTRACT(YEAR FROM scheduled_arrival_time) = 2024;
SELECT * FROM flight
```

flight_id	flight_number	airline_id	departure_airport_id	arrival_airport_id	scheduled_departure_time	scheduled_arrival_time
1	FL1	1	12	37	2027-08-31 11:00:54.484217	2027-04-27 22:00:00
2	FL2	1	12	37	2026-10-08 18:39:42.595192	2026-10-24 20:20:00
3	FL3	1	12	37	2027-07-15 18:54:30.500813	2027-09-28 06:10:00
4	FL4	1	12	37	2027-07-16 19:32:22.111321	2026-12-10 22:10:00
5	FL5	1	12	37	2027-04-24 19:05:50.722743	2027-01-10 07:00:00
6	FL6	1	12	37	2027-09-07 16:37:29.456909	2026-10-21 17:30:00
7	FL7	1	12	37	2027-01-22 20:19:44.818849	2026-11-15 10:30:00

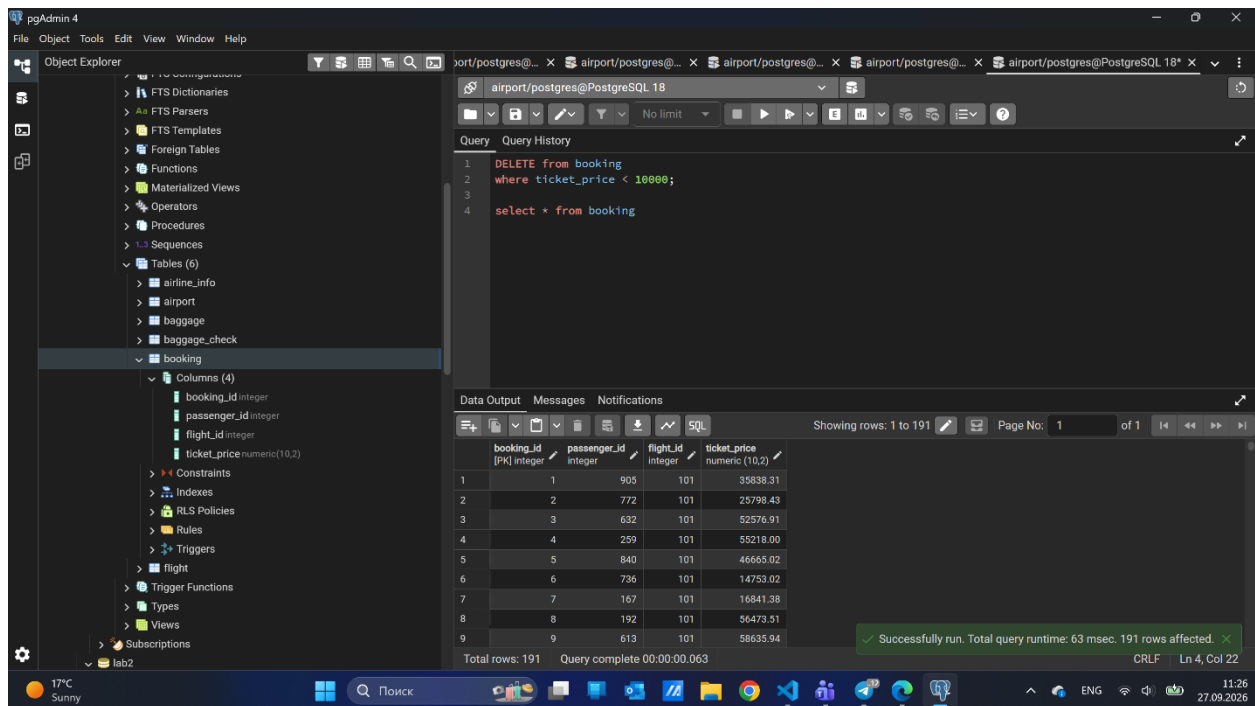
6. Increase the price of all tickets in booking table for flights by 15%.

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including the 'booking' table. The main pane shows a SQL query to increase the ticket price by 15%. The 'Data Output' tab displays the results of the query, showing 7 rows of booking data. A green 'Successfully' message is visible in the bottom right corner.

```
UPDATE booking
SET ticket_price = ticket_price * 1.15;
SELECT * FROM booking
```

booking_id	passenger_id	flight_id	ticket_price
1	905	101	35838.31
2	772	101	25798.43
3	632	101	52576.91
4	259	101	55218.00
5	840	101	46665.02
6	736	101	14753.02
7	167	101	16841.38

7. Delete all tickets where price is less than 10000.



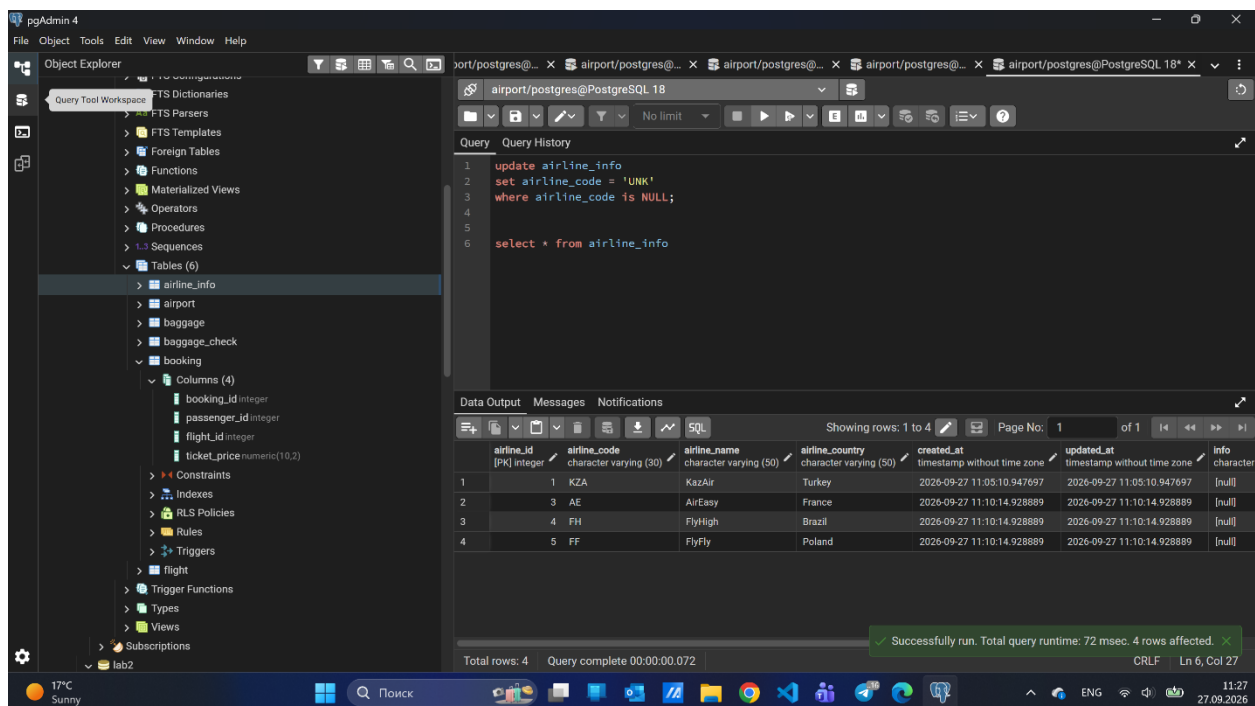
The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, with the 'booking' table selected under the 'airline' database. The right pane shows a SQL query window with the following query:

```
1 DELETE from booking
2 where ticket_price < 10000;
3
4 select * from booking
```

The 'Data Output' pane shows the result of the query, displaying 191 rows of data from the 'booking' table. The columns are: booking_id (PK) integer, passenger_id integer, flight_id integer, and ticket_price numeric(10,2). The status bar indicates 'Successfully run. Total query runtime: 63 msec. 191 rows affected.'

booking_id	passenger_id	flight_id	ticket_price
1	1	905	35838.31
2	2	772	25798.43
3	3	632	52576.91
4	4	259	55218.00
5	5	840	46665.02
6	6	736	14753.02
7	7	167	16841.38
8	8	192	56473.51
9	9	613	58635.94

8. Set a default code 'UNK' (Unknown) for any airline in the airline table where airline_code is missing (NULL).



The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, with the 'airline_info' table selected under the 'airline' database. The right pane shows a SQL query window with the following query:

```
1 update airline_info
2 set airline_code = 'UNK'
3 where airline_code is NULL;
4
5
6 select * from airline_info
```

The 'Data Output' pane shows the result of the query, displaying 4 rows of data from the 'airline_info' table. The columns are: airline_id (PK) integer, airline_code character varying(30), airline_name character varying(50), airline_country character varying(50), created_at timestamp without time zone, updated_at timestamp without time zone, and info character. The status bar indicates 'Successfully run. Total query runtime: 72 msec. 4 rows affected.'

airline_id	airline_code	airline_name	airline_country	created_at	updated_at	info
1	KZA	KazAir	Turkey	2026-09-27 11:05:10.947697	2026-09-27 11:05:10.947697	[null]
2	AE	AirEasy	France	2026-09-27 11:10:14.928889	2026-09-27 11:10:14.928889	[null]
3	FH	FlyHigh	Brazil	2026-09-27 11:10:14.928889	2026-09-27 11:10:14.928889	[null]
4	FF	FlyFly	Poland	2026-09-27 11:10:14.928889	2026-09-27 11:10:14.928889	[null]

9. Delete all baggage check records created before June 1, 2023, whose result is still 'Not checked'.

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, including the 'baggage_check' table. The central pane contains a SQL query:

```
1 DELETE FROM baggage_check
2 WHERE created_at < '2023-06-01'
3 and check_result = 'Not checked';
4
5 select * from baggage_check;
6
```

The bottom pane shows the 'Data Output' tab with a table of results. The table has columns: baggage_check_id, check_result, created_at, updated_at, booking_id, and passenger_id. The data shows 200 rows, with the first 9 rows displayed.

baggage_check_id	check_result	created_at	updated_at	booking_id	passenger_id
1	Checked	2026-03-14 04:27:19.030212	2026-09-27 11:28:17.676828	60	575
2	Not checked	2026-07-13 08:04:21.19822	2026-09-27 11:28:17.676828	60	64
3	Checked	2025-10-05 19:58:56.703582	2026-09-27 11:28:17.676828	60	270
4	Checked	2025-10-31 10:50:18.390416	2026-09-27 11:28:17.676828	60	818
5	Not checked	2026-02-05 18:50:04.913589	2026-09-27 11:28:17.676828	60	230
6	Not checked	2026-09-14 13:08:51.261042	2026-09-27 11:28:17.676828	60	65
7	Not checked	2026-06-05 02:45:06.991899	2026-09-27 11:28:17.676828	60	67
8	Not checked	2026-06-22 16:40:05.263528	2026-09-27 11:28:17.676828	60	441
9	Not checked	2025-10-15 04:58:45.371257	2026-09-27 11:28:17.676828	60	116

Total rows: 200 Query complete 00:00:01.989

10. Delete all entries from the airport table where state is missing (NULL) and the city is either 'Mlawe' or 'Kepuh'.

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, including the 'airport' table. The central pane contains a SQL query:

```
1 delete from airport
2 where state is NULL
3 and city in ('Mlawe', 'Kepuh');
4
5 select * from airport;
```

The bottom pane shows the 'Data Output' tab with a table of results. The table has columns: airport_id, airport_name, country, state, city, created_at, and updated_at. The data shows 200 rows, with the first 8 rows displayed.

airport_id	airport_name	country	state	city	created_at	updated_at
1	Airport_1	Country_42	State_41	City_16	2026-09-27 11:01:41.607537	2026-09-27 11
2	Airport_2	Country_18	State_15	City_30	2026-09-27 11:01:41.607537	2026-09-27 11
3	Airport_3	Country_6	State_0	City_41	2026-09-27 11:01:41.607537	2026-09-27 11
4	Airport_4	Country_5	State_48	City_39	2026-09-27 11:01:41.607537	2026-09-27 11
5	Airport_5	Country_30	State_19	City_30	2026-09-27 11:01:41.607537	2026-09-27 11
6	Airport_6	Country_21	State_7	City_5	2026-09-27 11:01:41.607537	2026-09-27 11
7	Airport_7	Country_34	State_13	City_34	2026-09-27 11:01:41.607537	2026-09-27 11
8	Airport_8	Country_26	State_19	City_42	2026-09-27 11:01:41.607537	2026-09-27 11

Total rows: 200 Query complete 00:00:00.080

11. Insert a new unchecked baggage inspection record and immediately return its newly generated baggage_check_id and created_at timestamp.

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, with the 'baggage' table selected. The right pane shows the SQL query editor with the following query:

```
1 insert into baggage_check(check_result, created_at, updated_at)
2 values ('Not checked', NOW(), NOW())
3 returning baggage_check_id, created_at;
```

The Data Output pane shows the result of the query:

baggage_check_id [PK] integer	created_at timestamp without time zone
201	2026-09-27 11:33:43.192887

A green status bar at the bottom indicates: "Successfully run. Total query runtime: 73 msec. 1 rows affected."

12. Convert all country names in the airline table to uppercase to ensure data consistency.

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure, with the 'airline_info' table selected. The right pane shows the SQL query editor with the following query:

```
1 update airline_info
2 set airline_name = UPPER(airline_name);
3
4 select * from airline_info
```

The Data Output pane shows the result of the query:

airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	info character	
1	1	KZA	KAZAIR	Turkey	2026-09-27 11:05:10.947697	2026-09-27 11:05:10.947697	[null]
2	3	AE	AIREASY	France	2026-09-27 11:10:14.928889	2026-09-27 11:10:14.928889	[null]
3	4	FH	FLYHIGH	Brazil	2026-09-27 11:10:14.928889	2026-09-27 11:10:14.928889	[null]
4	5	FF	FLYFLY	Poland	2026-09-27 11:10:14.928889	2026-09-27 11:10:14.928889	[null]

A green status bar at the bottom indicates: "Query complete 00:00:00.070"

13. Update both the name and country for the airline with airline_id = 5. Set the name to 'Global Airways' and the country to 'United Kingdom', while updating update_at to the current date.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including the 'airline_info' table. The main pane shows a SQL query being executed:

```
1 update airline_info
2 set airline_name = 'Global Airways',
3     airline_country = 'United Kingdom',
4     updated_at = NOW()
5 where airline_id = 5;
6
7 select * from airline_info
```

The Data Output pane shows the results of the query, displaying the updated record for airline_id 5:

airline_id	airline_code	airline_name	airline_country	created_at	updated_at	info
1	KZA	KAZAIR	Turkey	2026-09-27 11:05:10.947697	2026-09-27 11:05:10.947697	[null]
2	AE	AIREASY	France	2026-09-27 11:10:14.928889	2026-09-27 11:10:14.928889	[null]
3	FH	FLYHIGH	Brazil	2026-09-27 11:10:14.928889	2026-09-27 11:10:14.928889	[null]
4	FF	Global Airways	United Kingdom	2026-09-27 11:10:14.928889	2026-09-27 11:36:48.474932	[null]

14. Set the state field to 'Capital District' for airports located in specific cities: 'Astana', 'London', and 'Tokyo'

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including the 'airport' table. The main pane shows a SQL query being executed:

```
1 update airport
2 set state = 'Capital District'
3 where city in ('Astana', 'Tokyo', 'London');
4
5 select * from airport
```

The Data Output pane shows the results of the query, displaying the updated records for airports in Astana, London, and Tokyo:

airport_id	airport_name	country	state	city	created_at	updated_at
196	Airport_196	Country_9	State_0	City_26	2026-09-27 11:01:41.607537	2026-09-27 11:01:41.607537
197	Airport_197	Country_12	State_39	City_44	2026-09-27 11:01:41.607537	2026-09-27 11:01:41.607537
198	Airport_198	Country_35	State_32	City_21	2026-09-27 11:01:41.607537	2026-09-27 11:01:41.607537
199	Airport_199	Country_12	State_6	City_21	2026-09-27 11:01:41.607537	2026-09-27 11:01:41.607537
200	Airport_200	Country_16	State_38	City_45	2026-09-27 11:01:41.607537	2026-09-27 11:01:41.607537
201	Astana airport	Kazakhstan	Capital District	Astana	2026-09-27 11:39:53.283771	2026-09-27 11:39:53.283771
202	London airport	United Kingdom	Capital District	London	2026-09-27 11:39:53.283771	2026-09-27 11:39:53.283771
203	Tokyo airport	Japan	Capital District	Tokyo	2026-09-27 11:39:53.283771	2026-09-27 11:39:53.283771

15. Mark all baggage check records (baggage_check) created in March 2024 with a status of 'Not checked' as completed ('Checked').

The screenshot displays the pgAdmin 4 interface. On the left, the Object Explorer shows the database structure, including the 'baggage_check' table. The central pane shows a SQL query being executed:

```
1 update baggage_check
2 set check_result = 'Checked'
3 where created_at >= '2024-03-01'
4 and created_at <= '2024-03-31';
5
6
7 select * from baggage_check
```

The bottom pane shows the query results in a table format. The table has 9 columns: baggage_check_id, check_result, created_at, updated_at, booking_id, and passenger_id. The results show 9 rows of data, with the 'check_result' column updated to 'Checked' for records created in March 2024.

baggage_check_id	check_result	created_at	updated_at	booking_id	passenger_id
1	Checked	2026-03-14 04:27:19.030212	2026-09-27 11:28:17.676828	60	575
2	Not checked	2026-07-13 08:04:21.19822	2026-09-27 11:28:17.676828	60	64
3	Checked	2025-10-05 19:58:56.703582	2026-09-27 11:28:17.676828	60	270
4	Checked	2025-10-31 10:50:18.390416	2026-09-27 11:28:17.676828	60	818
5	Not checked	2026-02-05 18:50:04.913589	2026-09-27 11:28:17.676828	60	230
6	Not checked	2026-09-14 13:08:51.261042	2026-09-27 11:28:17.676828	60	65
7	Not checked	2026-06-05 02:45:06.991899	2026-09-27 11:28:17.676828	60	67
8	Not checked	2026-06-22 16:40:05.263528	2026-09-27 11:28:17.676828	60	441
9	Not checked	2025-10-15 04:58:45.371257	2026-09-27 11:28:17.676828	60	116

Total rows: 201 Query complete 00:00:00.320